

High Energy & Particle Physics

Worked Solutions

Introduction

This document gives worked solutions to the *High Energy & Particle Physics* problem sheet supplied with the January 2020 problem sheet.

The calculations and conservation-law checks follow the supplied solution document. In particular, the particle-physics notation used here follows the sheet: baryon number is $1/3$ for a quark and $-1/3$ for an antiquark, lepton number is $+1$ for leptons and -1 for antileptons, and antiparticles have the same mass but opposite additive quantum numbers.

1 Question 1 — Particle interactions and conservation laws

1.1 (i) Beta-minus decay

At the quark level,

$$d \rightarrow u + W^{-},$$

followed by

$$W^{-} \rightarrow e^{-} + \bar{\nu}_e.$$

At the first vertex:

	Q/e	B	L
d	$-1/3$	$1/3$	0
u	$+2/3$	$1/3$	0
W^{-}	-1	0	0

Hence

$$-\frac{1}{3} = \frac{2}{3} - 1, \quad \frac{1}{3} = \frac{1}{3}, \quad 0 = 0 + 0.$$

At the second vertex,

$$W^{-} \rightarrow e^{-} + \bar{\nu}_e,$$

so

$$-1 = -1 + 0,$$

and

$$L(W^{-}) = 0 = 1 + (-1).$$

Thus charge, baryon number and lepton number are all conserved at both vertices.

1.2 (ii) Beta-plus decay

The corresponding process is

$$u \rightarrow d + W^+,$$

followed by

$$W^+ \rightarrow e^+ + \nu_e.$$

At the first vertex,

$$+\frac{2}{3} = -\frac{1}{3} + 1, \quad \frac{1}{3} = \frac{1}{3} + 0, \quad 0 = 0 + 0.$$

At the second vertex,

$$+1 = +1 + 0,$$

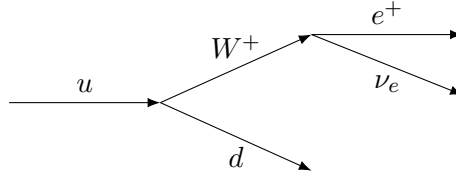
and

$$0 = (-1) + 1.$$

Therefore

charge, baryon number and lepton number are conserved at every vertex.

A corresponding Feynman-diagram sketch is:



1.3 (iii) $\pi^- p \rightarrow n + \pi^- + \pi^+$

Using

$$\pi^- = \bar{u}d, \quad p = uud, \quad n = udd, \quad \pi^+ = u\bar{d},$$

the interaction is

$$\bar{u}d + uud \rightarrow udd + \bar{u}d + u\bar{d}.$$

For charge,

$$Q_{\text{initial}}/e = -1 + 1 = 0,$$

while

$$Q_{\text{final}}/e = 0 - 1 + 1 = 0.$$

For baryon number,

$$B_{\text{initial}} = 0 + 1 = 1,$$

and

$$B_{\text{final}} = 1 + 0 + 0 = 1.$$

Therefore both charge and baryon number are conserved.

1.4 (iv) $K^+ \rightarrow \pi^+ + \pi^0$

The charged kaon has quark content

$$K^+ = u\bar{s}.$$

The charged pion is

$$\pi^+ = u\bar{d},$$

while the neutral pion is a superposition of

$$u\bar{u} \quad \text{and} \quad d\bar{d}.$$

The initial charge is

$$Q(K^+) = +\frac{2}{3} + \frac{1}{3} = +1.$$

The final charge is

$$Q(\pi^+) + Q(\pi^0) = (+1) + 0 = +1.$$

Therefore

$$\boxed{Q_{\text{initial}} = Q_{\text{final}} = +e}.$$

1.5 (v) Quark binding energies

The supplied masses are

$$m_{K^+} = 0.52616m_p, \quad m_{\pi^+} = 0.14875m_p,$$

and

$$m_p c^2 = 938.27 \text{ MeV}.$$

For the kaon,

$$m_{K^+} c^2 = 0.52616(938.27) = 493.68 \text{ MeV}.$$

The constituent quark masses are

$$m_u c^2 = 2.4 \text{ MeV}, \quad m_s c^2 = 95 \text{ MeV}.$$

Define the binding contribution by

$$B = (m_u + m_s - m_{K^+}) c^2.$$

Then

$$B_{K^+} = 2.4 + 95 - 493.68 = -396.28 \text{ MeV}.$$

Hence

$$\boxed{B_{K^+} \approx -396.2 \text{ MeV}}.$$

The negative sign indicates that the hadron mass-energy is much lower than the sum of the quoted bare quark masses in this simplified constituent picture.

For the pion,

$$m_{\pi^+} c^2 = 0.14875(938.27) = 139.57 \text{ MeV}.$$

Therefore

$$B_{\pi^+} = 2.4 + 4.8 - 139.57 = -132.37 \text{ MeV},$$

so

$$\boxed{B_{\pi^+} \approx -132.4 \text{ MeV}}.$$

These should not be interpreted as ordinary potential energies of two free quarks: quarks are confined, and the simple subtraction is only the model requested by the problem.

1.6 (vi) $\Sigma^- \rightarrow n + e^- + \bar{\nu}_e$

The Σ^- has quark content

$$\Sigma^- = dds,$$

while

$$n = udd.$$

At quark level, one strange quark changes through the weak interaction:

$$s \rightarrow u + W^-,$$

followed by

$$W^- \rightarrow e^- + \bar{\nu}_e.$$

At the first vertex,

$$-\frac{1}{3} = \frac{2}{3} - 1,$$

$$\frac{1}{3} = \frac{1}{3} + 0,$$

and lepton number remains zero.

At the second vertex,

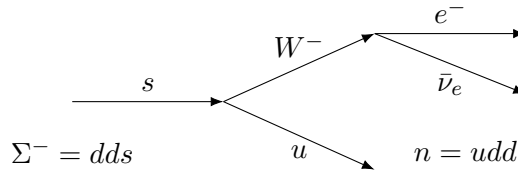
$$-1 = -1 + 0$$

and

$$0 = 1 - 1.$$

Thus all three relevant additive conservation laws hold at each vertex.

The quark-level decay can be sketched as



1.7 (vii) Cloud and bubble chambers

A **cloud chamber** contains a supersaturated vapour. A charged high-energy particle passing through the chamber ionises molecules along its path. The ions act as nucleation sites, so droplets condense along the trajectory and form a visible track.

A **bubble chamber** contains a liquid maintained in a superheated state. A charged particle ionises the liquid along its path, providing nucleation sites for bubbles. The bubbles grow along the trajectory and form a visible track.

A magnetic field can be used with either technique to curve charged-particle tracks, allowing the momentum and charge sign to be inferred.

1.8 (viii) Pair production of muons

To create a muon–antimuon pair,

$$\gamma \rightarrow \mu^- + \mu^+,$$

the photon must supply at least the two rest-mass energies:

$$E_{\gamma, \min} = 2m_{\mu}c^2.$$

With

$$m_\mu c^2 = 105.66 \text{ MeV},$$

we obtain

$$E_{\gamma,\min} = 2(105.66) = 211.32 \text{ MeV}.$$

Using

$$1 \text{ GeV} = 1.6 \times 10^{-10} \text{ J},$$

or equivalently

$$1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J},$$

gives

$$E_{\gamma,\min} \approx 211.32(1.6 \times 10^{-13}) = 3.38 \times 10^{-11} \text{ J}.$$

Thus

$$\boxed{E_{\gamma,\min} \approx 3.38 \times 10^{-11} \text{ J}}.$$

The minimum wavelength follows from

$$E = \frac{hc}{\lambda},$$

so

$$\lambda_{\max} = \frac{hc}{E_{\gamma,\min}} \approx 5.88 \times 10^{-15} \text{ m}.$$

Therefore

$$\boxed{\lambda_{\max} \approx 5.9 \text{ fm}}.$$

At threshold the pair is produced with negligible kinetic energy in the idealised threshold calculation.

1.9 (ix) Electron–positron pair moving at $0.8c$

The total energy of one electron or positron is

$$E = \gamma mc^2,$$

where

$$\gamma = \frac{1}{\sqrt{1 - u^2/c^2}}.$$

For

$$u = 0.8c,$$

$$\gamma = \frac{1}{\sqrt{1 - 0.8^2}} = \frac{5}{3}.$$

The photon must supply the total energy of both particles:

$$E_\gamma = 2\gamma m_e c^2.$$

Since

$$m_e c^2 = 0.51100 \text{ MeV},$$

we find

$$E_\gamma = 2 \left(\frac{5}{3} \right) (0.51100) \approx 1.70 \text{ MeV}.$$

Hence

$$\boxed{E_\gamma \approx 1.70 \text{ MeV}}.$$

1.10 (x) Bremsstrahlung shower

A high-energy electron entering matter is strongly deflected by the electric fields of atomic nuclei. Because it is accelerating, it emits bremsstrahlung photons.

A sufficiently energetic photon can then undergo pair production:

$$\gamma \rightarrow e^- + e^+.$$

The newly created electron and positron are themselves deflected by nuclei and can emit further bremsstrahlung photons. Those photons can again produce pairs.

Schematically,

$$e^- \rightarrow e^- + \gamma \rightarrow e^- + e^- + e^+ \rightarrow \text{many } e^\pm + \gamma \rightarrow \text{particle shower}.$$

The repeated branching produces a cascade of particles. The resulting shower can therefore provide an observable signature of the original high-energy electron.

1.11 (xi) Koide mass ratio for u, d, s

The given relation is

$$\frac{m_u + m_d + m_s}{(\sqrt{m_u} + \sqrt{m_d} + \sqrt{m_s})^2} \approx \frac{k}{9}.$$

Using

$$m_u = 2.4, \quad m_d = 4.8, \quad m_s = 95$$

in MeV/c^2 ,

$$R_q = \frac{2.4 + 4.8 + 95}{(\sqrt{2.4} + \sqrt{4.8} + \sqrt{95})^2}.$$

Numerically,

$$R_q \approx 0.562.$$

Therefore

$$k \approx 9(0.562) \approx 5.06.$$

The requested integer is

$$\boxed{k \approx 5}.$$

2 Question 2 — Strangeness

2.1 (i) Determining the quark content of K^0

The reaction is

$$K^- + p \rightarrow \Omega^- + K^+ + K^0.$$

The Ω^- has quark content

$$\Omega^- = sss$$

and hence strangeness

$$S = -3.$$

The initial K^- has

$$K^- = \bar{u}s,$$

so

$$S_{\text{initial}} = -1.$$

The proton has

$$S = 0.$$

The final K^+ has

$$K^+ = u\bar{s}$$

and therefore

$$S = +1.$$

For strangeness to be conserved, the neutral kaon must supply

$$-1 = (-3) + (+1) + S(K^0),$$

giving

$$S(K^0) = +1.$$

A neutral meson with strangeness +1 must contain an anti-strange quark:

$$K^0 = u\bar{s}.$$

Indeed,

$$Q(K^0) = +\frac{2}{3} + \frac{1}{3} = +1$$

would not be neutral if interpreted this way, so the neutral state is instead

$$\boxed{K^0 = d\bar{s}},$$

for which

$$Q = -\frac{1}{3} + \frac{1}{3} = 0, \quad S = +1, \quad B = 0.$$

Thus

$$\boxed{K^0 = d\bar{s}}.$$

2.2 (ii) $\Omega^- \rightarrow \Xi^0 + \pi^-$

The quark compositions are

$$\Omega^- = sss, \quad \Xi^0 = uss, \quad \pi^- = d\bar{u}.$$

The weak process can be represented by

$$s \rightarrow u + W^-,$$

followed by

$$W^- \rightarrow d + \bar{u}.$$

The first vertex conserves charge:

$$-\frac{1}{3} = \frac{2}{3} - 1,$$

and baryon number:

$$\frac{1}{3} = \frac{1}{3} + 0.$$

The second vertex conserves charge:

$$-1 = -\frac{1}{3} - \frac{2}{3},$$

and baryon number:

$$0 = \frac{1}{3} - \frac{1}{3}.$$

However, strangeness changes:

$$S(\Omega^-) = -3,$$

while

$$S(\Xi^0) = -2, \quad S(\pi^-) = 0.$$

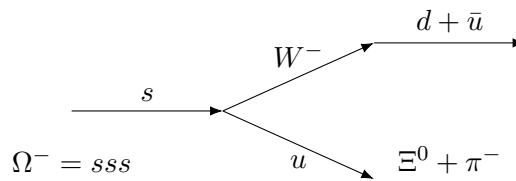
Thus

$$-3 \neq -2.$$

Therefore

strangeness is not conserved in this weak decay,

A quark-level sketch is



as expected for a weak interaction.

3 Question 3 — Tau decay

3.1 (i) Feynman diagram

The decay

$$\tau^- \rightarrow \pi^- + \nu_\tau$$

occurs through

$$\tau^- \rightarrow W^- + \nu_\tau,$$

followed by

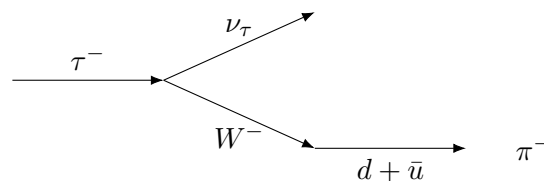
$$W^- \rightarrow d + \bar{u},$$

which hadronise to form the π^- .

Thus, schematically,

$$\tau^- \longrightarrow \nu_\tau + W^-, \quad W^- \longrightarrow d + \bar{u} \longrightarrow \pi^-.$$

A Feynman-diagram sketch is



3.2 (ii) Why is the tau neutrino required?

The tau lepton has lepton number

$$L_\tau = +1.$$

The W^- boson has

$$L_\tau = 0.$$

Therefore the vertex

$$\tau^- \rightarrow W^- + \nu_\tau$$

requires

$$1 = 0 + 1.$$

Without the tau neutrino, tau-lepton number would change from +1 to 0.

Hence

tau lepton number would not be conserved without ν_τ .

3.3 (iii) Kinetic energy of the tau

The tau has

$$m_\tau c^2 = 1776.86 \text{ MeV}$$

and speed

$$u = 0.2c.$$

Its Lorentz factor is

$$\gamma = \frac{1}{\sqrt{1 - 0.2^2}} \approx 1.02062.$$

The total energy is

$$E_\tau = \gamma m_\tau c^2.$$

The kinetic energy is

$$K_\tau = (\gamma - 1)m_\tau c^2.$$

Therefore

$$K_\tau = (1.02062 - 1)(1776.86) \approx 36.64 \text{ MeV}.$$

Thus

$K_\tau \approx 36.6 \text{ MeV}$.

3.4 (iv) Energy of the tau neutrino

The pion has speed

$$u_\pi = 0.99c.$$

Its mass is given by

$$m_\pi c^2 = 0.14875 m_p c^2 = 0.14875(938.27) \approx 139.57 \text{ MeV}.$$

Its Lorentz factor is

$$\gamma_\pi = \frac{1}{\sqrt{1 - 0.99^2}} \approx 7.0888.$$

Therefore

$$E_\pi = \gamma_\pi m_\pi c^2 \approx 989.37 \text{ MeV}.$$

The tau total energy is

$$E_\tau = \gamma_\tau m_\tau c^2 \approx 1813.50 \text{ MeV}.$$

Assuming the tau neutrino is effectively massless,

$$E_{\nu_\tau} = E_\tau - E_\pi.$$

Hence

$$E_{\nu_\tau} \approx 1813.50 - 989.37 = 824.13 \text{ MeV}.$$

Thus

$$\boxed{E_{\nu_\tau} \approx 824.1 \text{ MeV}}.$$

3.5 (v) Relativistic energy–momentum invariant

The relativistic momentum is

$$p = \gamma mv = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} mv.$$

The total energy is

$$E = \gamma mc^2.$$

Therefore

$$E^2 - p^2 c^2 = \gamma^2 m^2 c^4 - \gamma^2 m^2 v^2 c^2.$$

Factoring,

$$E^2 - p^2 c^2 = \gamma^2 m^2 c^4 \left(1 - \frac{v^2}{c^2}\right).$$

Since

$$\gamma^2 = \frac{1}{1 - v^2/c^2},$$

we obtain

$$\boxed{E^2 - p^2 c^2 = m^2 c^4}.$$

For an effectively massless tau neutrino,

$$m_{\nu_\tau} \approx 0,$$

so

$$E_\nu^2 - p_\nu^2 c^2 = 0.$$

Hence

$$E_\nu = p_\nu c,$$

or

$$p_\nu = \frac{E_\nu}{c}.$$

Using

$$E_\nu = 824.13 \text{ MeV},$$

$$\boxed{p_\nu \approx 824.1 \text{ MeV}/c}.$$

In SI units,

$$p_\nu = \frac{824.13 \times 10^6 (1.602 \times 10^{-19})}{2.998 \times 10^8} \approx 4.40 \times 10^{-19} \text{ kg m s}^{-1}.$$

Thus

$$\boxed{p_\nu \approx 4.40 \times 10^{-19} \text{ kg m s}^{-1}}.$$

3.6 (vi) Koide formulae

For the charged leptons,

$$R_\ell = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}.$$

Using

$$m_e = 0.51100, \quad m_\mu = 105.66, \quad m_\tau = 1776.86$$

in MeV/c^2 ,

$$R_\ell \approx 0.66666.$$

Thus

$$\boxed{R_\ell \approx \frac{2}{3}}.$$

For the quarks,

$$R_q = \frac{m_c + m_b + m_t}{(\sqrt{m_c} + \sqrt{m_b} + \sqrt{m_t})^2}.$$

Using

$$m_c = 1275, \quad m_b = 4180, \quad m_t = 172440$$

in MeV/c^2 ,

$$R_q \approx 0.669.$$

Thus

$$\boxed{R_q \approx \frac{2}{3}}.$$

The numerical closeness of both expressions to $2/3$ is the empirical pattern highlighted by the problem.

4 Question 4 — Compton scattering

Consider an incident photon of wavelength λ scattering through angle θ from an initially stationary electron.

The Compton wavelength shift is

$$\boxed{\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)}.$$

The following derivation obtains this result from relativistic energy–momentum conservation.

4.1 (i) Photon momentum and the de Broglie relation

For any particle,

$$E^2 - p^2 c^2 = m^2 c^4.$$

For a photon,

$$m = 0,$$

so

$$E^2 = p^2 c^2.$$

Since $E > 0$,

$$E = pc.$$

Planck's relation gives

$$E = hf = \frac{hc}{\lambda}.$$

Therefore

$$pc = \frac{hc}{\lambda},$$

and hence

$$\boxed{p = \frac{h}{\lambda}}.$$

This is the photon form of the de Broglie relation.

4.2 (ii) Momentum conservation

Before the collision, the photon momentum is

$$p_{\lambda} = \frac{h}{\lambda}.$$

After scattering,

$$p_{\lambda'} = \frac{h}{\lambda'}.$$

Momentum conservation gives

$$\vec{p}_{\lambda} = \vec{p}_{\lambda'} + \vec{p}_e.$$

Therefore

$$\vec{p}_e = \vec{p}_{\lambda} - \vec{p}_{\lambda'}.$$

Taking the magnitude and applying the cosine rule,

$$\boxed{p_e^2 = p_{\lambda}^2 + p_{\lambda'}^2 - 2p_{\lambda}p_{\lambda'} \cos \theta}.$$

4.3 (iii) Electron momentum from energy conservation

Initially the electron is at rest, so its energy is

$$E_{e,i} = m_e c^2.$$

Energy conservation gives

$$m_e c^2 + p_{\lambda} c = E_e + p_{\lambda'} c.$$

Hence

$$E_e = m_e c^2 + c(p_{\lambda} - p_{\lambda'}).$$

For the recoiling electron,

$$E_e^2 - p_e^2 c^2 = m_e^2 c^4.$$

Substitute the expression for E_e :

$$p_e^2 c^2 = [m_e c^2 + c(p_{\lambda} - p_{\lambda'})]^2 - m_e^2 c^4.$$

Expanding and dividing by c^2 ,

$$p_e^2 = (p_{\lambda} - p_{\lambda'})^2 + 2m_e c(p_{\lambda} - p_{\lambda'}).$$

Thus

$$\boxed{p_e^2 = (p_{\lambda} - p_{\lambda'})^2 + 2m_e c(p_{\lambda} - p_{\lambda'})}.$$

4.4 (iv) Derivation of the Compton shift

Equating the two expressions for p_e^2 gives

$$p_\lambda^2 + p_{\lambda'}^2 - 2p_\lambda p_{\lambda'} \cos \theta = (p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'}).$$

Since

$$(p_\lambda - p_{\lambda'})^2 = p_\lambda^2 + p_{\lambda'}^2 - 2p_\lambda p_{\lambda'},$$

we obtain

$$2p_\lambda p_{\lambda'}(1 - \cos \theta) = 2m_e c(p_\lambda - p_{\lambda'}).$$

Therefore

$$p_\lambda p_{\lambda'}(1 - \cos \theta) = m_e c(p_\lambda - p_{\lambda'}).$$

Using

$$p_\lambda = \frac{h}{\lambda}, \quad p_{\lambda'} = \frac{h}{\lambda'},$$

gives

$$\frac{h^2}{\lambda \lambda'}(1 - \cos \theta) = m_e c \left(\frac{h}{\lambda} - \frac{h}{\lambda'} \right).$$

Multiplying through by $\lambda \lambda' / h$,

$$\frac{h}{m_e c}(1 - \cos \theta) = \lambda' - \lambda.$$

Thus

$$\boxed{\Delta \lambda = \lambda' - \lambda = \frac{h}{m_e c}(1 - \cos \theta)}.$$

4.5 (v) Maximum wavelength shift and a 10% maximum-shift photon

The shift is greatest when

$$1 - \cos \theta$$

is greatest. Since

$$-1 \leq \cos \theta \leq 1,$$

the maximum occurs at

$$\boxed{\theta = 180^\circ}.$$

Thus

$$\Delta \lambda_{\max} = \frac{2h}{m_e c}.$$

The problem asks for an incident X-ray whose fractional wavelength shift is 10% of this maximum:

$$\frac{\Delta \lambda_{\max}}{\lambda} = 0.1.$$

Therefore

$$\frac{2h}{m_e c \lambda} = 0.1.$$

Using

$$E_\gamma = \frac{hc}{\lambda},$$

we have

$$\frac{\Delta \lambda_{\max}}{\lambda} = \frac{2E_\gamma}{m_e c^2}.$$

Hence

$$0.1 = \frac{2E_\gamma}{m_e c^2},$$

so

$$E_\gamma = 0.05 m_e c^2.$$

With

$$m_e c^2 = 511 \text{ keV},$$

$$E_\gamma = 0.05(511) = 25.55 \text{ keV}.$$

Thus

$$\boxed{E_\gamma \approx 25.6 \text{ keV}}.$$

4.6 (vi) Electron recoil angle and speed

Resolve momentum conservation into components.

Taking the incident photon direction as the x -axis,

$$p_\lambda = p_{\lambda'} \cos \theta + p_e \cos \phi,$$

and

$$0 = p_{\lambda'} \sin \theta - p_e \sin \phi.$$

Therefore

$$p_e \sin \phi = p_{\lambda'} \sin \theta,$$

and

$$p_e \cos \phi = p_\lambda - p_{\lambda'} \cos \theta.$$

Dividing,

$$\tan \phi = \frac{p_{\lambda'} \sin \theta}{p_\lambda - p_{\lambda'} \cos \theta}.$$

Since

$$\frac{p_\lambda}{p_{\lambda'}} = \frac{\lambda'}{\lambda},$$

and

$$\frac{\lambda'}{\lambda} = 1 + \frac{h}{m_e c \lambda} (1 - \cos \theta),$$

we obtain

$$\boxed{\tan \phi = \frac{\sin \theta}{1 + \frac{h}{m_e c \lambda} (1 - \cos \theta) - \cos \theta}}.$$

The electron momentum can be written as

$$\boxed{p_e = \sqrt{(p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'})}}.$$

Using

$$p_\lambda = \frac{h}{\lambda}, \quad p_{\lambda'} = \frac{h}{\lambda'},$$

and

$$\lambda' = \lambda + \frac{h}{m_e c} (1 - \cos \theta),$$

this gives p_e entirely in terms of λ , θ and the constants.

The relativistic momentum relation is

$$p_e = \gamma m_e v = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} m_e v.$$

Solving for v ,

$$v = c \left[\left(\frac{m_e c}{p_e} \right)^2 + 1 \right]^{-1/2}.$$

4.7 (vii) Maximum electron recoil speed

The maximum wavelength shift occurs for

$$\theta = 180^\circ.$$

At this angle,

$$\lambda' = \lambda + \frac{2h}{m_e c}.$$

The scattered photon therefore has the minimum possible momentum

$$p_{\lambda'} = \frac{h}{\lambda'},$$

so the difference

$$p_\lambda - p_{\lambda'}$$

is maximised. From

$$p_e^2 = (p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'}),$$

the electron momentum is consequently maximised.

Energy conservation gives

$$\frac{hc}{\lambda} + m_e c^2 = \frac{hc}{\lambda'} + \gamma m_e c^2.$$

Thus

$$\gamma = 1 + \frac{h}{m_e c} \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right).$$

Since

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}},$$

we obtain

$$v_{\max} = c \sqrt{1 - \left[\frac{m_e c^2}{m_e c^2 + \frac{hc}{\lambda} - \frac{hc}{\lambda'}} \right]^2},$$

with

$$\lambda' = \lambda + \frac{2h}{m_e c}.$$

4.8 (viii) Computational plots

The question supplies two requested plots: the fractional wavelength shift $\Delta\lambda/\lambda$ and the electron recoil speed v/c , each as a function of the photon scattering angle θ , for incident photon energies 50, 100, 200, 500, and 1000 keV. The code below reproduces those quantities and the two plots.

For an incident photon energy E ,

$$\lambda = \frac{hc}{E}, \quad \frac{\Delta\lambda}{\lambda} = \frac{E}{m_e c^2} (1 - \cos \theta).$$

The scattered wavelength is

$$\lambda' = \lambda + \frac{h}{m_e c} (1 - \cos \theta),$$

so

$$p_\lambda = \frac{h}{\lambda}, \quad p_{\lambda'} = \frac{h}{\lambda'}.$$

The recoil momentum and speed are

$$p_e = \sqrt{(p_\lambda - p_{\lambda'})^2 + 2m_e c(p_\lambda - p_{\lambda'})},$$

$$\frac{v}{c} = \left[1 + \left(\frac{m_e c}{p_e} \right)^2 \right]^{-1/2}.$$

A program reproducing the two requested graphs is:

```
import numpy as np
import matplotlib.pyplot as plt

h = 6.62607015e-34
c = 2.99792458e8
me = 9.1093837e-31
e = 1.602176634e-19

E_keV = [50, 100, 200, 500, 1000]
theta = np.linspace(1e-8, np.pi, 1000)

fig1, ax1 = plt.subplots()
fig2, ax2 = plt.subplots()

for Ek in E_keV:
    E = Ek * 1e3 * e
    lam = h*c/E
    lam_p = lam + h/(me*c)*(1 - np.cos(theta))

    p = h/lam
    p_p = h/lam_p
    dp = p - p_p
    p_e = np.sqrt(dp**2 + 2*me*c*dp)

    frac = (lam_p - lam)/lam
    beta = 1/np.sqrt(1 + (me*c/p_e)**2)

    ax1.plot(np.degrees(theta), frac, label=f'E={Ek}keV')
    ax2.plot(np.degrees(theta), beta, label=f'E={Ek}keV')

for ax in (ax1, ax2):
    ax.set_xlabel('Photon_scattering_angle_theta_/deg')
```



```

ax.grid(True, alpha=0.25)
ax.legend()

ax1.set_ylabel('Delta_lambda/lambda')
ax1.set_title('Compton scattering of X-ray photon off an electron')
ax2.set_ylabel('Electron recoil speed v/c')
ax2.set_title('Compton scattering of X-ray photon off an electron')

fig1.tight_layout()
fig2.tight_layout()
fig1.savefig('compton_shift.png', dpi=180)
fig2.savefig('compton_speed.png', dpi=180)
plt.show()

```

The resulting plots are:

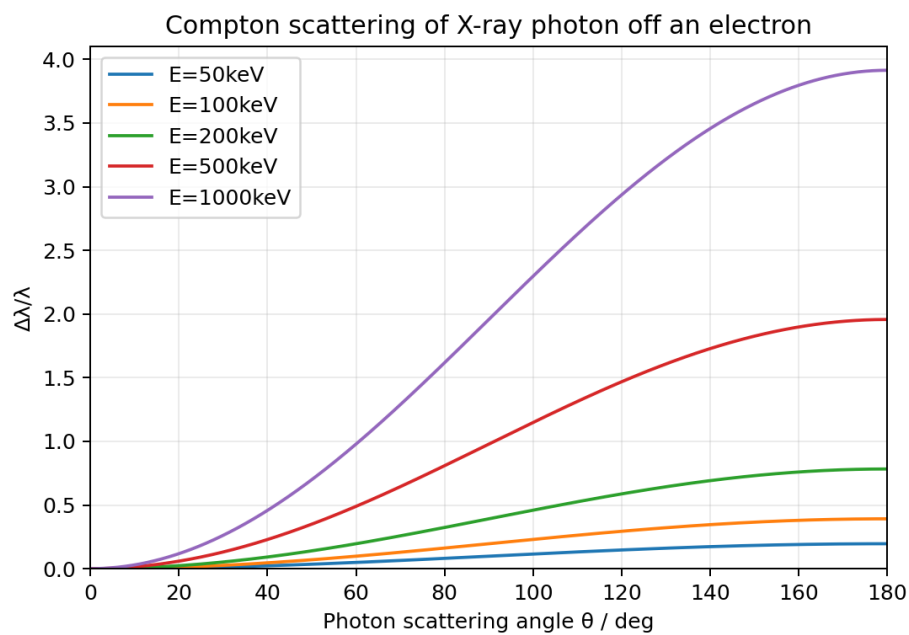


Figure 1: Fractional Compton wavelength shift for the five incident X-ray energies specified in the question.

Summary

The particle-physics sheet connects several central ideas:

- conservation of charge, baryon number, lepton number and, where appropriate, strangeness;
- quark-level descriptions of weak decays;
- relativistic energy and momentum;
- pair production and particle showers;
- the empirical Koide mass relation;
- the energy-momentum invariant

$$E^2 - p^2c^2 = m^2c^4;$$

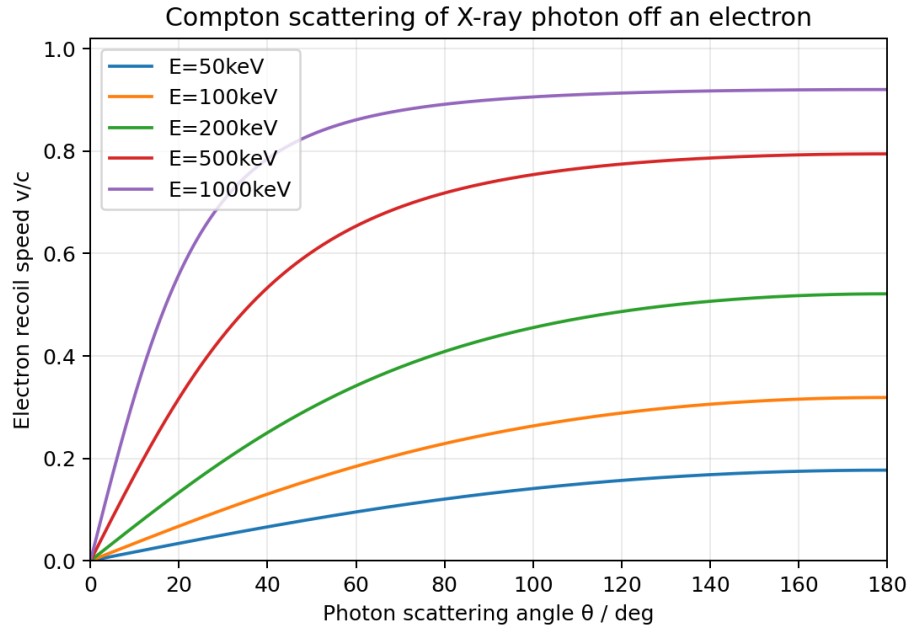


Figure 2: Electron recoil speed as a fraction of c for the five incident X-ray energies.

- and the particle description of light through Compton scattering.

The Compton-scattering problem in particular leads naturally to the computational model requested in the final part of the sheet: once the incident photon energy is specified, the wavelength shift, recoil momentum, recoil speed and recoil angle can all be evaluated numerically as functions of the scattering angle.